

Project title: Salary prediction Linear regression Project

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Dataset source :Kaggle –
Salary_dataset

Data Loading and Initial Inspection

- First, I uploaded the Salary dataset CSV dataset using the Google

Drive link. I used Python and Pandas to read the CSV file and store it in a DataFrame. After loading the dataset, I checked whether the data was loaded correctly.

- After loading the dataset, the first few rows were displayed using `df.head()` to understand the structure and contents

of the data `df.tail()` was used to display the last few rows of the dataset and verify that the data was loaded correctly.

- The `df.info()` function was then used to check the

number of rows and columns, column names, data types, and non-null values in the dataset.

- IS NULL FUNCTION

- checked the dataset for missing values using the `isnull()` Function.

```
0
YearsExperience  0
Salary          0
dtype: int64
```

- This helped me identify whether any columns contained missing or null values.

- I checked the number of missing values present in each column.

- Based on the result, I identified the columns that had missing values.
- Missing values were handled appropriately before continuing with further analysis.
 - I checked for duplicate records using the duplicated() function.
- There are no duplicates in my dataset

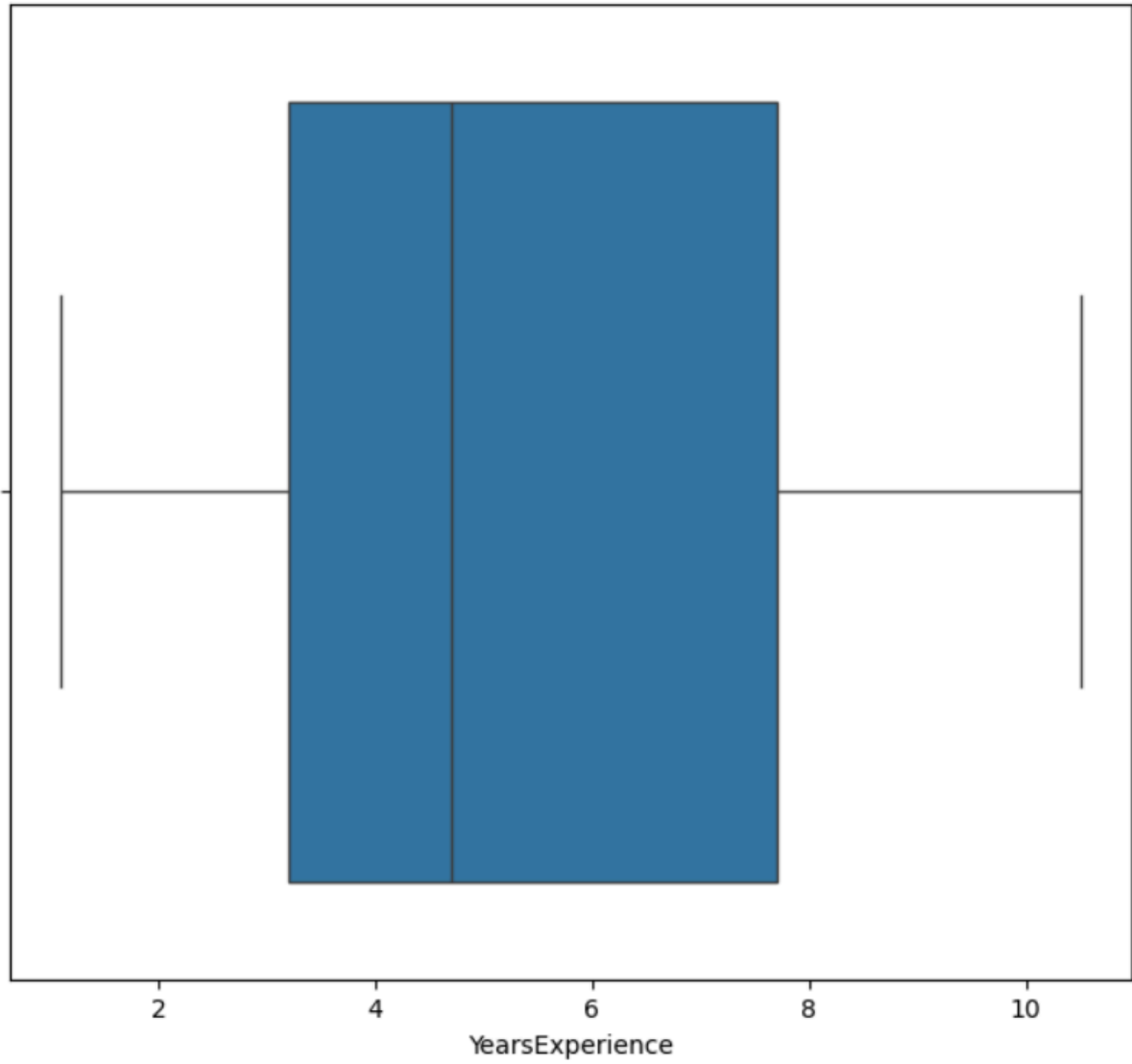
```
df.duplicated().sum()  
np.int64(0)
```

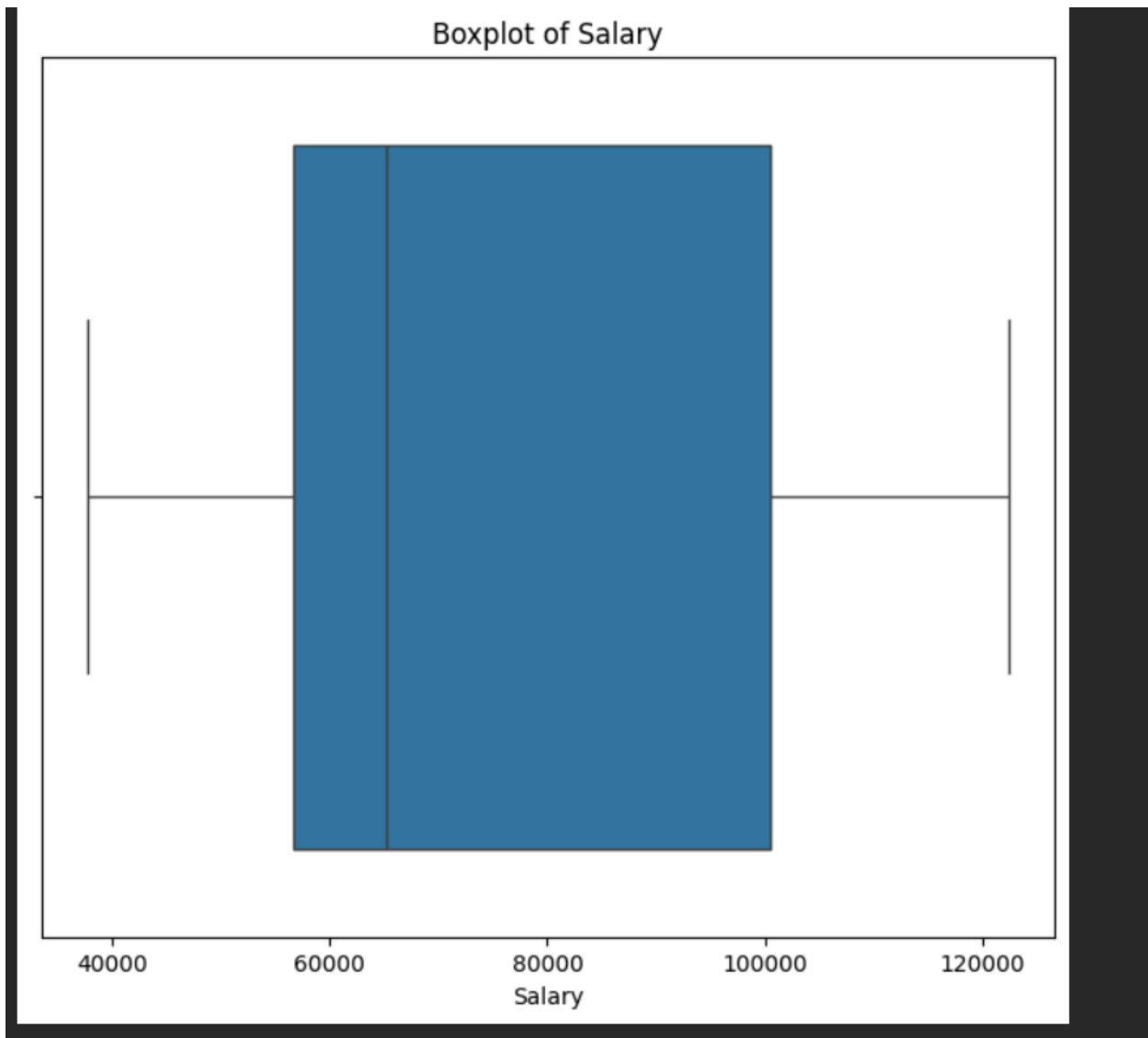
Box Plot Report:

- The box plot is used to identify the distribution of numerical data.

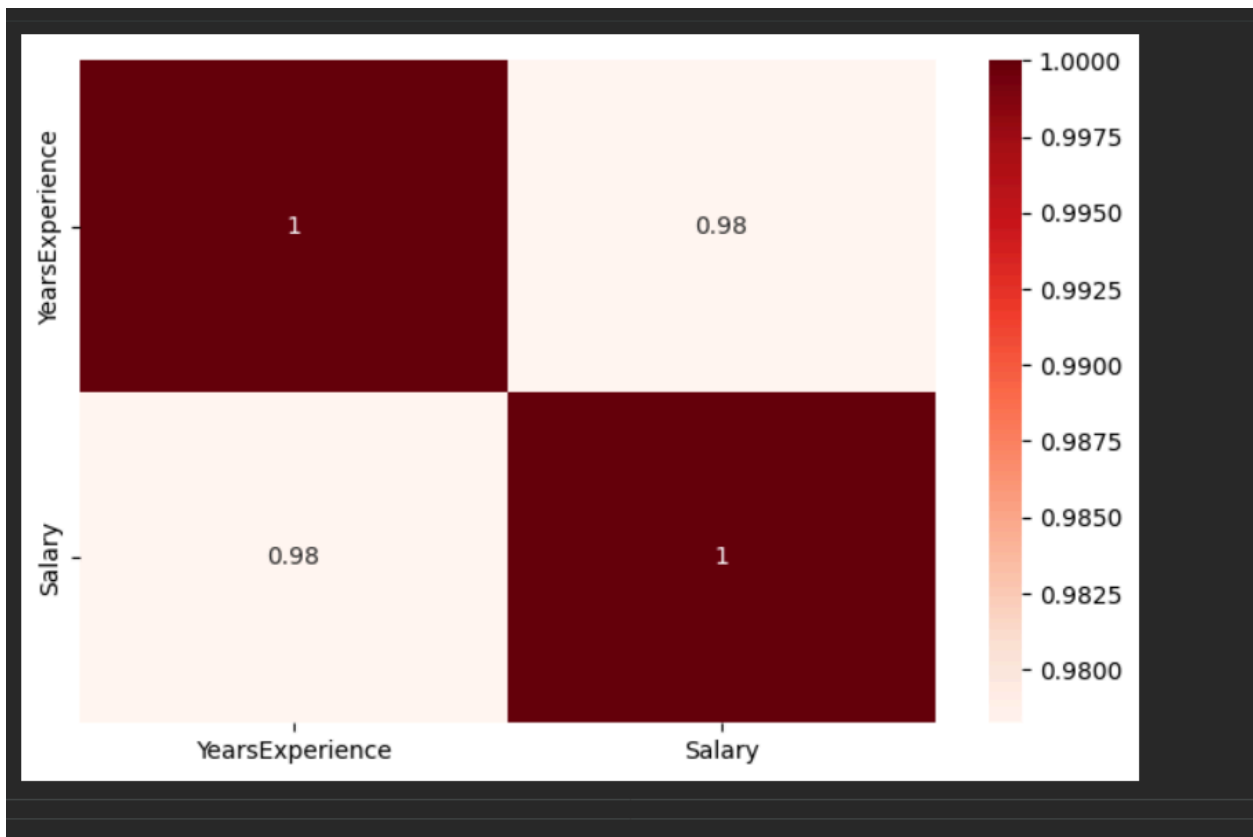
- It shows the median, lower and upper values, and possible outliers.
- The plot helps us understand how the values are spread.
- Any points outside the whiskers are considered possible outliers.
- This analysis helps us understand the variation in the numerical column.

Boxplot of YearsExperience





- Overall, this analysis helps us understand how rating and content type are related to year and Salary Prediction



1. Independent and Dependent Variables

The dataset was divided into **independent variables (X)** and a **dependent variable (Y)**.

- **Dependent Variable:** YearsExperience
- **Independent Variables:** All other features in the dataset.

The objective of the model is to predict the number of years of experience based on the given independent variables.

2. Train-Test Split

The dataset was divided into two subsets:

- **80% Training Data** – used to train the machine learning model.
- **20% Testing Data** – used to evaluate the model's performance on unseen data.

A fixed random state was used to ensure that the same data split can be reproduced whenever the model is executed.

3. Model Selection

Linear Regression was selected as the machine learning algorithm. Linear Regression is a supervised learning algorithm used to predict a continuous numerical target variable by identifying

the relationship between the independent variables and the dependent variable.

4. Model Training

The Linear Regression model was trained using the training dataset. During training, the model learned the relationship between the independent variables and **YearsExperience**.

5. Prediction

After training, the model was applied to the testing dataset to generate predicted values of **YearsExperience**. These predicted values were then compared with the actual values from the testing dataset.

6. Model Evaluation

The model's performance was evaluated using **Mean Squared Error (MSE)**. MSE measures the average squared difference between the actual and predicted values.

A lower MSE value indicates better prediction performance, as it means that the predicted values are closer to the actual values.

2. Evaluation Metrics

The following metrics were used:

- **Mean Absolute Error (MAE):**
Measures the average absolute difference between actual and

predicted values. A lower MAE indicates better performance.

- **R² Score:** Measures how well the independent variables explain the variation in the dependent variable. Its value ranges from 0 to 1, with a value closer to 1 indicating a better fit.

- **Root Mean Squared Error (RMSE):** Measures the square root of the average squared prediction error. A lower RMSE indicates better prediction accuracy.

3. Results

```
MAE: 0.5191320118982409  
r2: 0.9039484379486217  
RMSE: 0.5191320118982409
```

The model achieved an **R^2 score of approximately 0.904**, indicating that the model explains about **90.4% of the variation in YearsExperience** based on the independent variables.

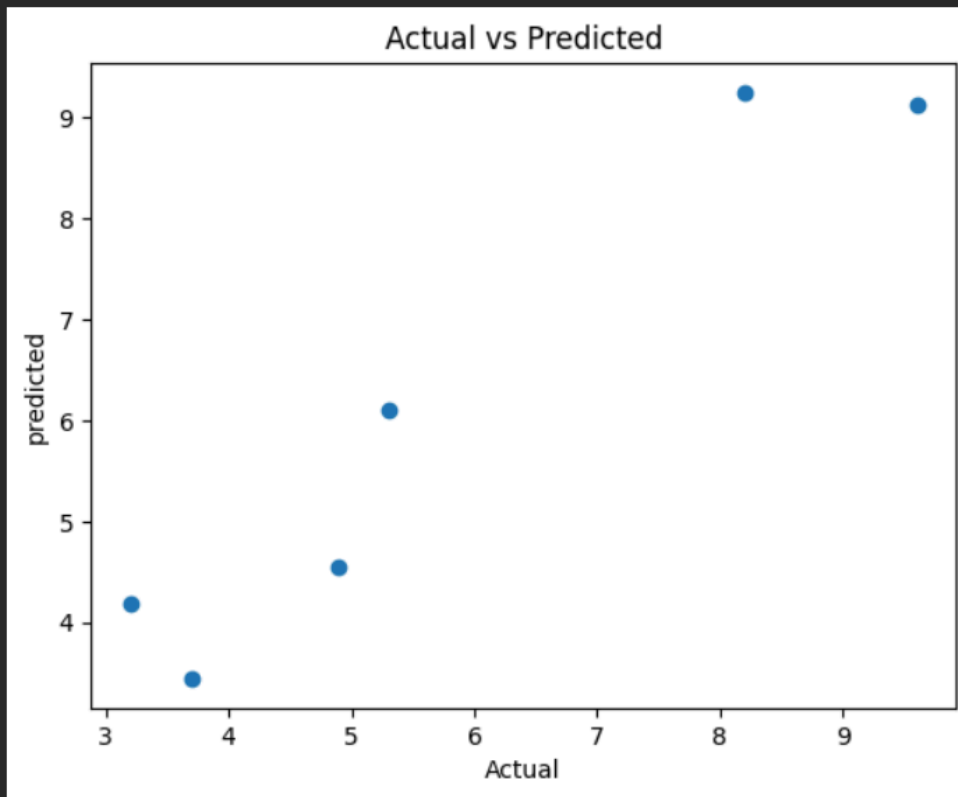
4. Actual vs Predicted Values

A comparison was made between the actual and predicted values. For example:

	Actual	Predicted
27	9.6	9.120593
15	4.9	4.545795
23	8.2	9.241061
17	5.3	6.096418
8	3.2	4.188282
9	3.7	3.445621

The error values are relatively low, showing that the model's predictions are reasonably close to the actual values.

This comparison demonstrates how closely the Linear Regression model predicts the actual **YearsExperience** values.



Actual vs Predicted Graph

A scatter plot was created to compare the **actual YearsExperience values** with the values predicted by the Linear Regression model.

- **X-axis:** Actual YearsExperience
- **Y-axis:** Predicted YearsExperience

- **Each point:** Represents one observation from the testing dataset.

The graph shows that the predicted values generally follow the actual values, indicating that the Linear Regression model is able to capture the relationship between the input features and YearsExperience.

Interpretation

The points are reasonably close to the expected **diagonal relationship between actual and predicted values**, which indicates good prediction performance. However, some differences can be observed

between the actual and predicted values for certain observations.

This visualization, together with the **R^2 score of 0.9039**, indicates that the model provides a good fit for the given dataset.

5. Conclusion

The Linear Regression model performed well on the testing dataset, achieving an **R^2 score of 0.9039**. The relatively low prediction error indicates that the model can effectively estimate YearsExperience from the given independent variables. Overall, the evaluation results show that Linear

Regression is a suitable model for this prediction task.